



Focus on
innovation.

We take care of the rest.

`baker` prints the board. `mixer` tests it. `proto`,
our AI engineer, walks you through every step.

`protopus-kappa.vercel.app`

01 · EXECUTIVE SUMMARY

A small electronics factory and a full test bench, on one desk

MORNING

You draw it

Proto checks the design and points at what will bite later.

MIDDAY

Baker builds it

Board printed, parts placed and soldered.

AFTERNOON

Mixer tests it

One cable, eight instruments, a reading you can trust.

EVENING

You try again

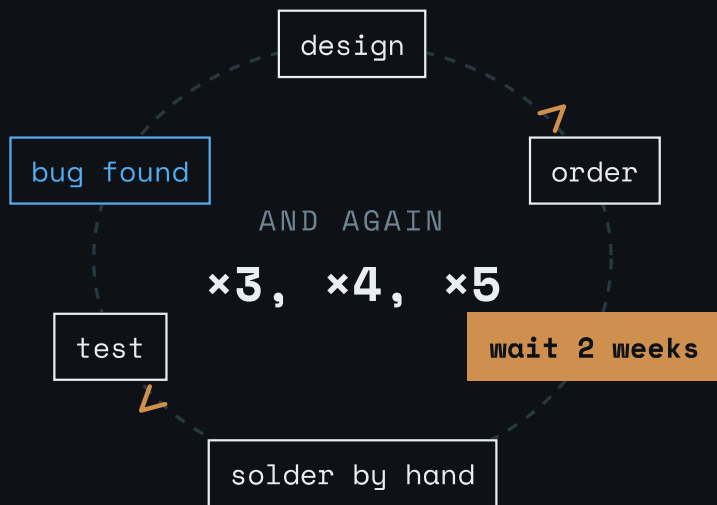
The next attempt is tonight, not two weeks from now.

WHAT MAKES IT
OURS

Making and testing on one desk, with Proto guiding every step instead of a hundred buttons.

02 · PROBLEM

A prototype is a loop, and every lap costs weeks



2

weeks waiting for a board to come back from a fabrication house

5

instruments to buy before you can even measure it

5

vendor apps and driver stacks to learn and keep alive

In my department a board ordered in week one gets measured in week four. **Nobody is short of ideas. They are short of tries.**

WHY NOW • compute is cheap enough for machines to carry the fidelity. People should carry the innovation.

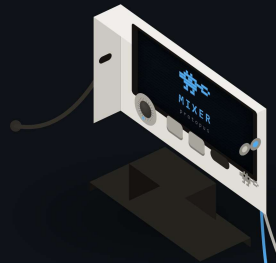
Two smart machines and one AI engineer



baker

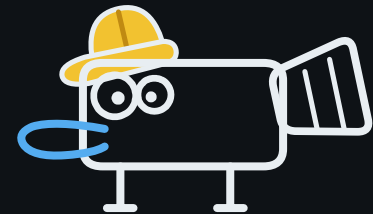
Prints PCBs with the components assembled.

Milling · drilling · mask · paste
· pick and place · reflow



mixer

Tests the board in real life.
Eight instruments behind one cable



proto

A platypus AI engineer inside both.

Flags your mistakes · guides every step

VISION · a bridge between idea and innovation for anyone who wants to bring theirs to the world.

DEMO 1

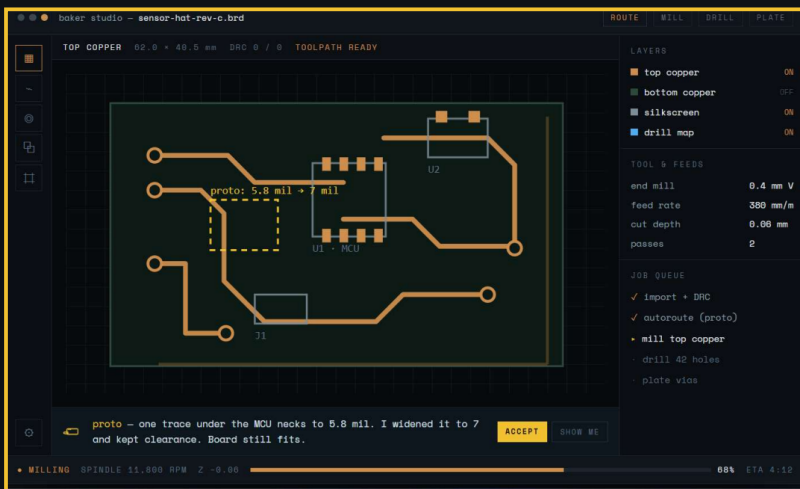
Website walk-through · protopus-kappa.vercel.app

DEMO 2

Video · raw Baker prototype running

04 · WHAT MAKES IT OURS

Proto: an AI engineer that understands your board



PROTOPUS Baker Studio

Proto has already found the thin trace, fixed it and planned the mill. You press Accept.

Probe matrix accessory

COMING NEXT

A grid of probes under the board gives Proto full access to every net on the PCB: it can measure anything, not just what you clip.

Simple enough that a **high-school student** can print a board.

01 READS THE BOARD

Understands schematics and layouts. Pulls designs in through the Flux.ai API.

02 MAKES IT

Generates the Gerbers and the Baker job: mill, drill, mask, stencil, placement.

03 DIAGNOSES IT

Runs the test through Mixer, understands the waveforms and says in plain words what is wrong.

04 HAS A CHARACTER

A platypus engineer with a personality developed through Gemini, guiding every step.

Between the DIY bench and industrial manufacturing

	DIY bench	protopus	industrial
	breadboards, hand etching, borrowed scopes	baker + mixer + proto	fab house, assembly house, bench
TIME PER LOOP	1-3 days	Same day	1-3 weeks + customs
EXPERTISE	A seasoned engineer	Proto guides you	Expert to design and test
ASSEMBLED	By hand, if you can	Yes, by Baker	Yes, at extra cost
TESTED	Only if you own a bench	Yes, by Mixer	Separate ₹ lakhs of instruments

The DIY bench needs an expert. The factory needs time and money.
Most ideas have neither.

Others solve one step. Nobody closes the whole loop.

PRODUCT	PRICE · ₹	MAKE	PLACE	TEST	GUIDE
Voltera V-One	~3.1 lakh	●	○	○	○
LPKF ProtoMat	8.6–24.5 lakh	●	○	○	○
Bantam Tools mill	~3.1 lakh	●	○	○	○
Digilent Analog Discovery 3	~33,000	○	○	●	○
JLCPCB, PCBWay	per order, weeks	●	●	○	○
protopus baker + mixer	3.34 lakh	●	●	●	●

List prices at ₹88 per US\$, before import duty. Voltera dispenses paste and reflows, but you place every part by hand.

voltera.io · Design World · Digilent

What we aim for: sell the machines, and one Proto plan runs all of them

PROJECTED PER UNIT · ₹ · 100-UNIT BATCHES	BAKER	MIXER
Hardware (target)	1,50,000	15,000
Assembly and QC	12,000	2,000
Packing and shipping	8,000	1,000
Warranty reserve	7,500	750
Software share	10,000	2,000
Sales and support	12,500	1,750
Projected total cost	2,00,000	22,500
Target selling price	2,99,000	34,999
Projected margin	~33%	~36%

Costed China-parts model: Baker ₹1,44,842 and Mixer ₹13,740 for hardware + assembly, inside these targets.

PLANNED · ONE PROTO PLAN · EVERY MACHINE

₹9,999 / year

One subscription per workspace: Baker Studio, Mixer Station, Proto and every update.

Consumables

Board blanks, paste and end mills, reordered as labs keep building.

Industrial variants, later

Bigger beds, multilayer and production test come after the base machines.

ROADMAP TO THESE NUMBERS

NOW	MONTHS 0-9 · THIS ASK	AFTER POC · SEED	100-UNIT BATCHES
Prototypes costed: Baker ₹3.73L, Mixer ₹74,847	₹16L: proof of concept, Baker Rev B, 5 production Mixers in studios	~₹1.5 Cr: China pilots, first 100 Mixers + 20 Bakers	The projected unit economics above

All unit costs, prices and margins are projections for production volumes, not current figures.

Start in the maker studios, then follow the people who watch them

₹6.4L cr global PCB market	₹3.1L cr global test and measurement	₹55K cr India PCB, ~15% a year	₹8.3K cr India test and measurement
33M+ active Arduino users worldwide	2.3M+ Hackster hardware community	10,000 Atal Tinkering Labs, 50,000 more planned	~2,500 Fab Labs in 125 countries

NOW

Maker studios

Free Baker and Mixer units, with a contract for featured videos. The Bambu Lab playbook.

NEXT

Hobbyists and startups

The people who watch those videos and lose weeks on every board.

ALONGSIDE

Prototyping labs

Tinkering labs, Fab Labs and college labs that already buy tools.

Where we are, told honestly

Baker

Raw prototype running on the bench. The video.

Software

Baker Studio with Proto inside. Website live.

Mixer

Idea phase. Architecture defined, not built.

SIGNAL

Met **Samuel Beek**, Founder & CEO of Schematik. Interested in funding us once the proof of concept is ready.

MONTH 0-2

Baker runs a full job, hands off

MONTH 2-4

First Mixer built and measuring

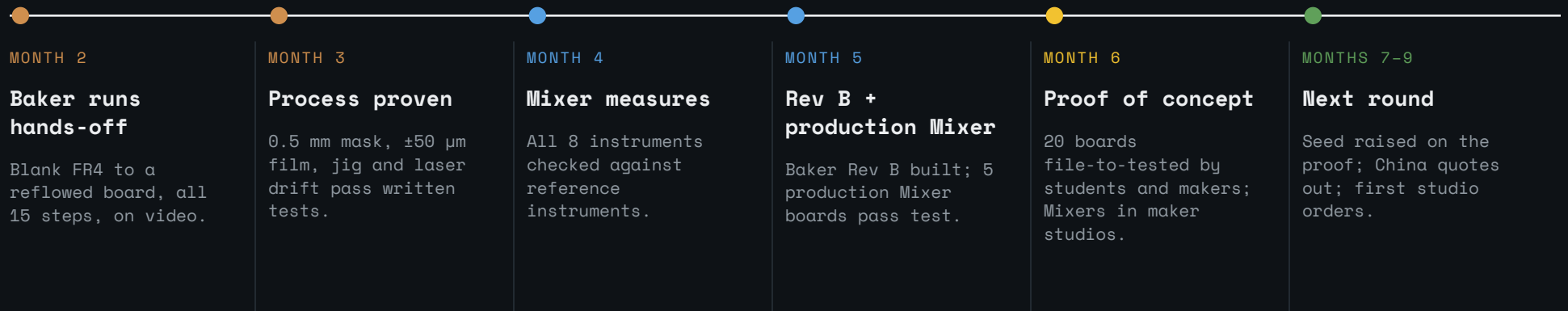
MONTH 4-6

Proof of concept, first maker studio units

AFTER POC

Next round, first sales

Our goal: a working proof of concept, one milestone you can watch at a time



₹16 lakh

the whole ask

9 months

6 to proof + 3 to raise

1 + 1

founder + a software engineer
who writes firmware

Every month

a demo video + Mixer test logs

Built in the university lab, tested with students and makers. Nothing else stands in the way.

₹16,00,000

sixteen lakh: nine months to a proof of concept, and time to raise.

₹4.48L	₹2.20L	₹4.50L	₹1.35L	₹1.50L	₹1.97L
Baker + Mixer builds, costed line by line	Baker Rev B, 5 production Mixer boards, consumables	One software + firmware engineer, nine months	Proto software and compute	Testing, tooling, lab safety	IP, outreach, 6.6% reserve
₹4,48,265	₹2,20,000	₹4,50,000	₹1,35,000	₹1,50,000	₹1,96,735

Enough to survive one failed iteration and to raise from strength. Nothing that belongs after the proof.

This builds the proof. The next round, at proof of concept, builds the business.

Where the ₹16 lakh goes, line by line

Baker	₹3,73,418
Frame & enclosure	22,795
Motion	80,382
Zone A: mill, drill, laser	38,270
Zone B: mask, UV, wipe	33,058
Stencil film	5,805
Placement + reflow	4,891
Board handling	300
Electronics & control	42,985
Resin printer (jigs)	77,698
First-run consumables	6,280
Parts	3,12,465
Import duty + courier (35%)	12,247
Contingency (15%)	48,707
Prototype total	₹3,73,418

Mixer	₹74,847
Capture + generation	18,198
Front end + instruments	15,752
Host, screen, I/O	26,752
Enclosure	1,510
Parts	62,212
Import duty + courier (35%)	2,872
Contingency (15%)	9,763
Prototype total	₹74,847

PRODUCTION UNIT, CHINA PARTS

Baker	₹1,44,842
target ₹1,50,000 · under by ₹5,158	
Mixer	₹13,740
target ₹15,000 · under by ₹1,260	

Use of funds	₹16,00,000
Baker prototype build	3,73,418
Mixer prototype build	74,847
Baker Rev B iteration	1,00,000
Mixer production board ×5	60,000
PoC consumables	60,000
Software + firmware engineer × 9 mo	4,50,000
Proto software + compute	1,35,000
Testing + tooling	1,00,000
Lab set-up + safety	50,000
IP + legal	75,000
Demos + studio outreach	50,000
Reserve (6.6% of non-hardware)	71,735
Total	16,00,000



People 28% · Hardware 28% · Iteration 14% · IP + reserve 12% · Lab + tooling 9% · Software 8%.

Basis: PROTOPUS cost report, 29 Sep 2026. 96 BOM lines, 81 priced from live listings. USD at ₹88; imported items carry 35% for duty and courier; 15% contingency on the subtotal. Salaries, iteration and other lines: PoC Budget sheet in the costing workbook.

After proof of concept: raise, build in China, run lean from Delhi

THE ROUND

~₹1.5 crore seed, at proof of concept

China pilots + first batches	₹70L
China set-up: fixtures, audits, IP	₹25L
Team of 5, twelve months	₹25L
Delhi centre + working capital	₹20L
Buffer	₹10L
Seed round	₹1.5 Cr

Pilot batches are sellable: 100 Mixers + 20 Bakers ~ ₹95L at list price. First conversation: Schematik, interested at PoC.

THE CHINA PLAN

Design in Delhi, build in Shenzhen

- Mixer: full turnkey from batch one. PCBA, box build, 100% test, packed.
- Baker: tested modules from China, final build in Delhi. Full turnkey once the factory price is ≤ \$1,305.
- 12 factories quoted → 2 paid pilots → 1 primary + 1 backup.
- EVT → DVT → PVT: 20 / 50 / 100 Mixers, 3 / 5 / 20 Bakers.

OPERATIONS

Delhi finalises, it doesn't build

- Every unit: scan, quick test, Proto activation, ship. A Mixer takes 5-10 minutes.
- Mixer is our production tester. We own the rig, the limits and the data.
- Hire: a mechanical and a manufacturing + quality engineer.
- IEC, customs broker and China trademarks before the first container.

MONTH 0-2

Design freeze, 12 RFQs out

MONTH 3-5

Pilots in China: EVT and DVT

MONTH 6-9

PVT: first 100 Mixers, 20 Bakers

MONTH 10-12

Mass production, first container to Delhi

Months counted from the round. Round sizing is indicative: pilots costed from the BOM, other lines estimated. Detail: PROTOPUS China plan.



Thank you.

Come and try the loop.

protopus-kappa.vercel.app

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MANAGEMENT TEAM & CORE SKILLS

Who is building this

Soham

Founder

Third year ECE, Faculty of Technology,
University of Delhi

Industrial design

Mechanical
architecture

Embedded electronics

Machine software

HIRING NEXT · a
software engineer who
also writes firmware.

